Minimizing Sample Redundancy for Label-Efficient Object Detection in Aerial Images

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***Abstract*— Objects in aerial images tend to be densely scattered and appear in arbitrary orientations, making the annotation process quite costly. To reduce the annotation cost, existing methods propose randomly annotating a proportion of images or objects for aerial object detection with fewer label usage. These approaches, however, can lead to redundancy in labels and inherit the biases associated with the imbalance in datasets. To minimize sample redundancy and alleviate data imbalance, we propose a novel labeling pattern that acquires heterogeneous object labels in a class-orthogonal manner, preserving a broader diversity of samples for each category with less annotation effort. To improve data utility, we design a dynamic multiview learning (DML) strategy to overcome the sample quantity-quality dilemma in current pseudo-labeling methods—a high pseudo-label threshold reduces sample quantity, while low thresholds compromise sample quality. First, DML separates model predictions into multiple hierarchies for finer screening, mitigating the suppression of unlabeled objects in binary pseudo-label (BPL) strategies. With this separation, DML learns to construct a new view by injecting high-quality samples and masking low-quality regions in this view, simultaneously expanding sample quantity while ensuring sample quality. Unlike previous methods that mine pseudo-labels solely from unlabeled regions, DML releases this constraint by learning to expand high-quality samples with a dynamic view. Extensive experiments on five benchmark datasets validate our method's state-of-the-art accuracy and label efficiency. Notably, with approximately 5% DOTA-v2.0 annotations, DML achieves nearly 90% of the fully supervised performance. The codes will be available at https://github.com/ZhangRuixiang-WHU/ALOD_DML/**

***Index Terms*— Aerial images, label-efficient object detection, sparse annotation.**

I. Introduction

AERIAL images, with large field of view and overhead perspectives, usually come with thousands of objects per image in diverse orientations, making them powerful tools for Earth observation. Detecting aerial objects with arbitrary orientations, as an important link toward intelligent Earth observation, has achieved significant progress propelled by fully supervised methods [1], [2], [3], [4], [5], [6], [7]. However, the aerial objects' inherent nature of large quantity, arbitrary orientation, and small size introduces great labor, time, and capital expenses for the object-by-object annotation under a fully supervised framework.

To relieve the labeling burden, a growing number of studies [8], [9], [10], [11] are focusing on *achieving or surpassing fully supervised performance with fewer label usage*. Although these methods have shown promising results, they typically employ a random labeling manner that overlooks the unique characteristics of aerial images, resulting in sample redundancy and class imbalance. As shown in Fig. 1, objects within the same aerial image share identical imaging properties (e.g., spatial resolution, tilt angle, and imaging time), resulting in similar geometric and appearance characteristics (e.g., scale, structure, and illumination). In contrast, objects from different images are captured under diverse imaging properties, leading to divergent representations. Meanwhile, the overhead view and uniform structure of objects in aerial images contribute to a high degree of similarity among objects of the same class. As a result, random labeling at the image level for label-efficient learning [e.g., semi-supervised object detection (SSOD)] will miss a lot of images with different representations. Instead, this full-image annotation method tends to retain more homogeneous samples in a single image, introducing sample redundancy and reducing sample efficiency. Although sparsely annotated object detection (SAOD) alleviates this issue by employing object-level random labeling, the unbalanced class distribution of objects in aerial imagery causes this approach to favor head categories with more frequent objects and suppress the representation of rare categories. This not only intensifies the intraclass sample redundancy but also inherits the class imbalance issue in aerial images.

To minimize sample redundancy and alleviate data imbalance, we present an attribute-aware labeling pattern for label-efficient object detection in aerial images (ALOD), aiming at promoting more data-efficient and class-unbiased learning. Our labeling pattern is simple, controllable, and feasible with three highlights: 1) label objects across the entire dataset to minimize domain gaps between images with differing imaging conditions; 2) define a class-orthogonal annotation basis for each image to reduce redundancy between similar intraclass objects and to avoid under-sampling of rare categories; and 3) randomly label K annotation basis per

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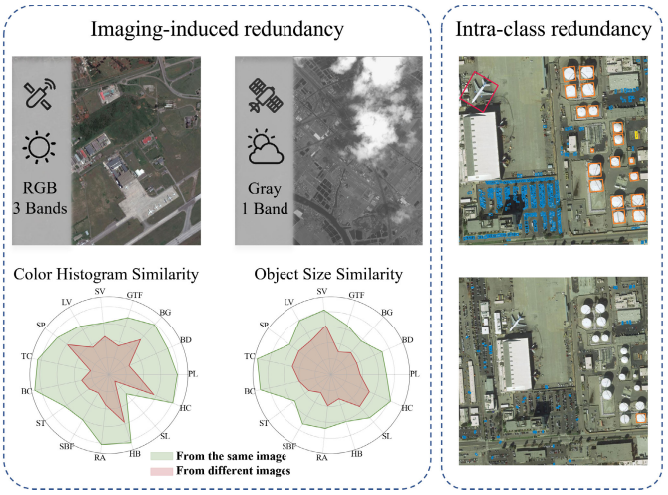


Fig. 1. *Imaging-induced redundancy:* Different imaging conditions (e.g., *satellite, weather, and spatial resolution*) have a significant impact on the representation of aerial images. Here are the characteristics of 1000 object pairs for each class from the same or different images in DOTA [1], including color histogram similarity and size similarity. As shown, objects within the same image exhibit higher similarity than objects from different images. Thus, randomly labeling images will lead to the loss of diverse image representations, retaining more homogeneous objects within the single image, and thus causing sample redundancy. *Intraclass redundancy:* The unbalanced distribution of objects in aerial imagery causes object-level random labeling to favor head classes with more frequent objects, further exacerbating sample redundancy.

image to control the labeling volume. While ALOD presents an effective labeling pattern, utilizing limited labels to achieve sufficient supervision remains a significant challenge. Current label-efficient methods [12], [13], [14] primarily mine pseudo-labels from unlabeled regions with a threshold to enrich the supervision. However, the model usually struggles to guarantee both sample quantity and sample quality: a higher pseudo-label threshold ensures label quality but misses many foreground samples, while a lower threshold increases label quantity but includes more uncertain samples. This dilemma further limits the mining and utilization of information from unlabeled regions.

To resolve this conflict and improve data utilization, we propose a dynamic multiview learning (DML) method that simultaneously improves both the quality and quantity of the samples. Instead of solely mining pseudo-labels from unlabeled regions, our method learns to expand high-quality samples by constructing a new view for model supervision, bypassing the sample quality-quantity dilemma. This is mainly realized via the proposed hierarchical pseudo-sample (HPS) learning. First of all, HPS separates model predictions into multiple hierarchies (i.e., high quality, uncertain, and low quality) for finer screening, mitigating the suppression of unlabeled objects in binary pseudo-label (BPL) strategies. After separation, HPS constructs a new view based on feedback from these predictions: HPS uses reliable samples (i.e., labels and high-quality predictions) to replace uncertain regions and mask low-quality areas in the new view, simultaneously expanding sample quantity while ensuring sample quality. Meanwhile, we propose a multilevel consistency (MLC) learning to jointly enforce the proximity of feature spaces and prediction spaces between multiple views. Feature alignment brings low-level structures and textures closer across view variants, while prediction alignment ensures the student model matches the teacher model's outputs. The joint alignment simultaneously brings close low-level features and high-level predictions, guaranteeing robust label-efficient learning.

Extensive experiments on five datasets from different scenarios demonstrate the DML's superiority over label-efficient aerial object detection counterparts in the following aspects: 1) *significant accuracy improvement:* the proposed learning pipeline significantly improves the baseline method on multiple datasets with limited label supervision; 2) *higher label efficiency:* with the same amount of training labels, DML outperforms both state-of-the-art SSOD methods and SAOD competitors by large margins; and 3) *strong generalization ability:* in addition to oriented object detection in aerial images, DML can also significantly improve the performance of oriented detection in other scenarios (e.g., SKU-110K-R) in a label-efficient manner.

The remainder of this article is organized as follows. First, we introduce related work on aerial object detection and label-efficient object detection in Section II. Next, we present the details of the designed labeling pattern ALOD and the learning method DML in Section III. Subsequently, we provide extensive experiments conducted on multiple datasets in Section IV. Additionally, we compare the strengths and weaknesses of several existing label-efficient methods in Section V. Finally, we conclude in Section VI.

II. Related Work

A. Oriented Object Detection

Oriented object detection differs significantly from traditional object detection in natural scenes due to the arbitrary orientations present in aerial images. This unique characteristic has led to the development of various approaches that focus on feature alignment, label assignment, and loss design tailored for oriented objects.

The evolution of oriented object detection has been marked by a trend toward more efficient architectures and simplified training strategies. In the early stages, the rotated region proposal network (RPN) [15] addressed the challenge of rotated object detection by introducing additional rotated anchor boxes. Subsequently, RoI Transformer [2] eliminated the need for rotated anchors by learning to transform RPN-generated horizontal proposals into their rotated counterparts. Building on this, Oriented R-CNN [4] refined the process by directly generating oriented proposals during the RPN stage, streamlining the detection pipeline.

Parallel to advancements in two-stage models, one-stage detection methods have also gained development. For instance, S2ANet [5] employs deformable convolutional networks (DCNs) to explicitly align features with anchors, improving detection accuracy. More recently, several studies [16], [17] have focused on customizing anchor-free detectors specifically for rotated object detection, further advancing the field.

While these methods have significantly improved the performance of oriented object detection, they depend heavily on a large volume of finely labeled samples, with rotated bounding

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boxes—a much more labor-intensive annotation process than standard bounding boxes. Consequently, label-efficient object detection has attracted growing interest in recent research.

B. Label-Efficient Object Detection

Label-efficient object detection methods have cast new insights into reducing the annotation burden of objects while maintaining comparable performance to the supervised one. Recent label-efficient object detection mainly revolves around weakly supervised object detection (WSOD), SSOD, and SAOD. WSOD aims to achieve object detection with minimal supervision, such as image-level annotations [18], [19], [20], [21], [22], [23], [24], [25], [26], point annotations [27], [28], [29], [30], or scribbles [31], [32], [33], without relying on precise bounding box annotations. Although these methods significantly reduce annotation costs, the absence of precise location coordinates often leads to suboptimal regression performance, making it challenging to meet the requirements for high-precision detection. In this work, we focus on reducing annotation costs by minimizing the number of required annotations, rather than relaxing the annotation format. This goal is primarily addressed through two approaches: SSOD and SAOD.

1) Semi-Supervised Object Detection: SSOD reduces annotation costs by utilizing a small set of fully labeled images combined with a large pool of unlabeled data. Given a dataset D with N images, SSOD randomly selects N_l images for complete annotation, leaving the remaining N_u images unannotated (see Fig. 2; SSOD). To address the challenges of leveraging unlabeled data, SSOD methods primarily employ pseudo-label learning [34] and consistency regularization [35]. Popular frameworks such as STAC [36], Unbiased Teacher [37], and Humble Teacher [38] integrate these strategies. Notably, Soft Teacher [39] adopts the Teacher–Student (T–S) framework from Mean-Teacher [40] for end-to-end training. Subsequent works have further refined SSOD by optimizing pseudo-label selection [41], [42], [43], [44], label assignment [44], [45], [46], [47], [48], [49], [50], and feature enhancement [51], [52], [53], [54], [55], [56]. Recently, SOOD [8] combines rotation-based consistency learning with the latest SSOD paradigms, achieving remarkable performance on aerial image datasets with densely arranged objects.

Despite these advancements, SSOD faces several challenges. Interimage discrepancy (Fig. 1), caused by distributional differences between labeled and unlabeled images—-often due to varying imaging conditions in aerial datasets–hinder effective pseudo-label generation. Additionally, complete annotations may include redundant labels for homogeneous objects, further complicating the learning process. These issues limit the effectiveness of SSOD in complex real-world scenarios, particularly in crowded scenes and when generalizing to unseen images.

2) Sparsely Annotated Object Detection: SAOD addresses label efficiency by sparsely annotating objects across each image in the dataset, ensuring the visibility of the entire dataset. Given a dataset D with N images, SAOD randomly annotates a few objects in each image (see Fig. 2, SAOD). This approach mitigates interimage discrepancy but introduces challenges due to incomplete annotations. To address the issue of missing annotations, which often cause foreground-background confusion during training, various methods have been proposed. Early approaches [57], [58], [59], [60] optimized network training by adjusting weights for missing samples. With the advancement of pseudo-label learning, methods like Co-mining [61] and SparseDet [12] employed twin networks to predict and generate pseudo-labels for missing instances. Calibrated Teacher [13] further improved pseudo-label learning by dynamically calibrating thresholds during training, following the Mean-Teacher framework [40]. Additionally, SAOD has been adapted to specific tasks, such as event camera detection [62], 3D object detection [63], [64], aerial image detection [14], and video object detection [65]. Notably, SIOD [66] introduced class-independent labeling and leveraged class similarity for pseudo-label generation, though its reliance on a specialized architecture limits its scalability and generalization for complex scenarios like aerial image detection.

Despite its flexibility, SAOD faces several challenges. First, the random annotation strategy commonly used in existing literature makes it difficult to control labeling ratios or determine which object to annotate without prior knowledge of the dataset. Second, random sampling treats images and objects independently, leading to redundancy in common classes and underrepresentation in rare classes. These issues indicate that current SAOD approaches may encounter limitations in operational feasibility and data distribution optimization when applied to aerial image object detection.

III. Approach

A. Labeling Pattern

To address the above problems in SSOD and SAOD, we propose the Attribute-aware label pattern for Label-efficient Object Detection. First, we employ sparse annotations for each image to avoid the interimage domain gap. Then, since similar objects within the same image often contain redundant information, annotating many objects of the same category can lead to sample redundancy. Therefore, we randomly annotate a small, equal number of K objects per category in each image (Fig. 2, ALOD shows the case of $K = 1$). Specifically, given an aerial image dataset D with N images, we label each image to get $D = \{I_i, G_i^k\}_{i=0}^N$, where G_i^k represents the sparse ground truth (GT) of K objects per category for the ith image. As a byproduct, this makes the labeling volume easy to control compared to randomly labeling a proportion of objects across the whole dataset. In the experimental section, we compare the direct supervision performance of these three labeling patterns under the same annotation ratio (see Section IV-C). Our results show that the ALOD labeling pattern provides better supervision and achieves superior detection performance with the same annotation proportion.

B. Problem Statement

In fully supervised training, an image I is divided into foreground and background regions, providing positive and negative samples for training. However, in our ALOD, the

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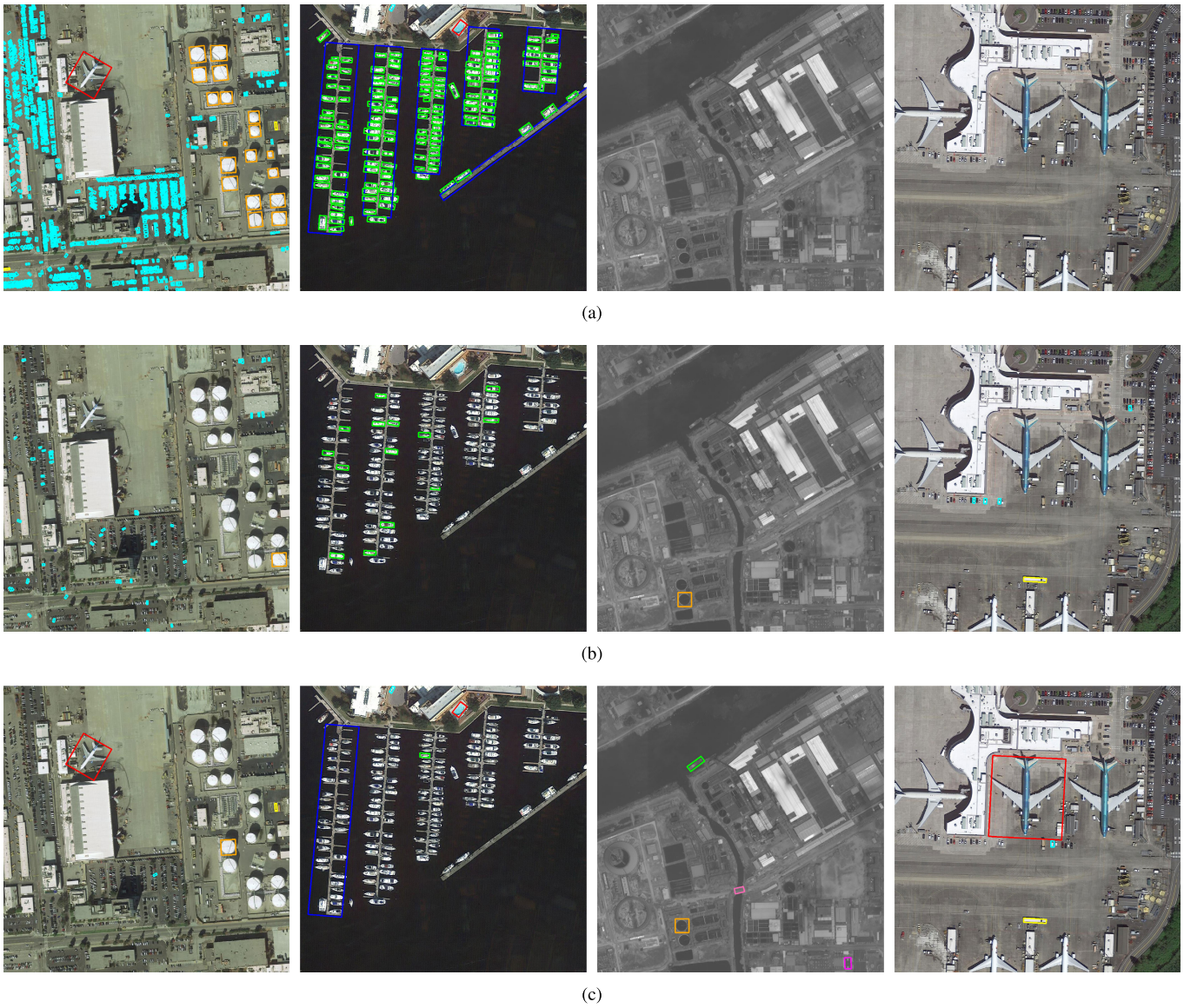


Fig. 2. Comparison of labeling patterns of several label-efficient methods. (a) SSOD: randomly label a few images, may miss samples from different domains. (b) SAOD: randomly label part objects, may miss samples from rare categories. (c) ALOD (ours): a balanced sampling of existing categories and all the domains.

image is separated into labeled and unlabeled regions, with the latter further divided into background and unlabeled foreground regions. This situation presents two challenges for network training: one is the shortage of positive samples due to the limited foreground annotations, and the other is the confusion in training, as unlabeled foreground objects may be wrongly classified as negative samples. Therefore, our goal is to improve label utility under limited labeling, supplementing positive examples and reducing confusion from misclassifications.

C. Pipeline Overview

1) T–S Architecture: Object detection with incomplete annotations is often solved with two main strategies: pseudo-label learning and consistency regularization. Combining these two strategies, earlier methods use forward propagation on heuristically augmented images to generate pseudo-labels for mutual supervision, as illustrated in Fig. 3(a). Subsequently, the growing adoption of the MeanTeacher [40], known for stabilizing training and reducing noise accumulation, inspires the development of more robust T–S architecture, as shown in Fig. 3(b). In this architecture, the Teacher network generates pseudo-labels from the weakly augmented view and updates through the exponential moving average (EMA) of the Student network, while the Student network is trained on the strongly augmented view using these pseudo-labels and sparse annotations.

Building on this T–S baseline, we propose a novel pipeline that simultaneously guarantees sample quantity and quality by constructing a multiview learning framework, as shown in Fig. 3(c). Specifically, we hierarchically separate the Teacher's predictions and feedback in the input space to create a new view. Additionally, we employ joint consistency learning in both the feature space and prediction space, further

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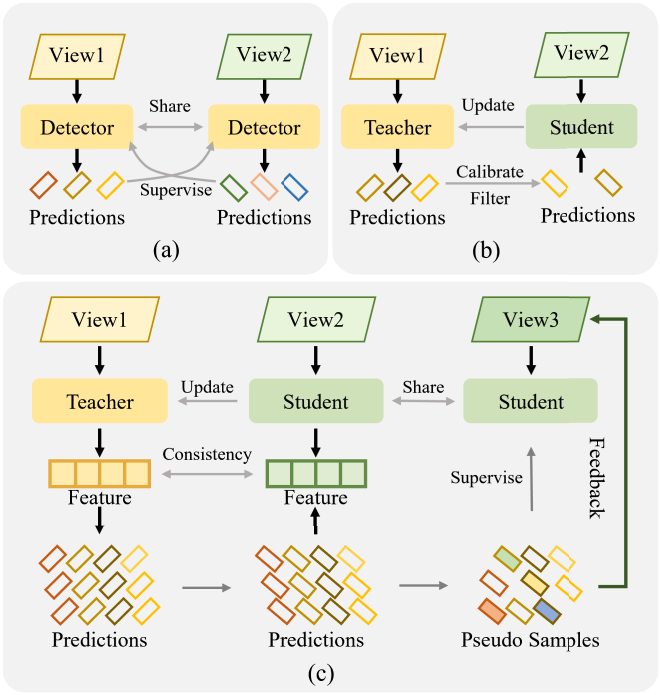


Fig. 3. Different pipelines for object detection with incomplete annotations. (a) and (b) Pipelines of some previous work. (c) Pipeline of the proposed DML.

strengthening the supervision signal and ensuring robust label-efficient learning.

2) BPL Learning → HPS Learning: Most previous methods employ pseudo-label learning following a binary approach, retaining only high-confidence predictions as pseudo-labels while discarding the rest of the predictions. This BPL learning strategy strictly treats all predictions other than high-score predictions as negative samples.

Formally, given an image I as input to the Teacher model $T(\cdot)$, the prediction set $P = T(I)$ is obtained, where $P = \{p_i | i = 1, 2, \ldots, N_p, p_i = (c_i, b_i, s_i)\}$. N_p is the number of predictions, p_i is the ith prediction, c_i, b_i, and s_i represent the class, the bounding box, and the score of this prediction, respectively. In BPL, the prediction set P is separated into positive and negative sets by

$$\begin{aligned} P_{\text{pos}} &= \{p_i \in P | s_i \geq \tau_{\text{high}}\} \\ P_{\text{neg}} &= \{p_i \in P | s_i < \tau_{\text{high}}\} \end{aligned} \tag{1}$$

where τ_{high} is the threshold for high-quality predictions. Then, we can get the pseudo-label set Y_p of image I by $Y_p = P_{\text{pos}}$.

However, in sparsely annotated aerial imagery, the detector is often confused by unlabeled objects since most of them are treated as negative objects during training. This results in a scarcity of high-quality predictions and a considerable number of ambiguous predictions ($s_i < \tau_{\text{high}}$). BPL thus faces the quality–quantity dilemma—a high τ_{high} leads to a lack of pseudo-labels, while a low τ_{high} compromises the quality of pseudo-labels, ultimately affecting the detector's performance.

To mitigate this issue, we propose a novel approach called HPS learning. Unlike BPL, HPS does not simply discard ambiguous predictions. Instead, these ambiguous predictions ($s_i < \tau_{\text{high}}$) are fed back into the image space for pseudo-sample replacement or masking, rather than being labeled as background. This approach allows the network to incorporate more nuanced supervision, enriching the training data with information from these ambiguous predictions. In HPS, we separate the predictions into three sets (high-quality predictions P_{high}, uncertain predictions P_{unc}, and low-quality predictions P_{low}) by

$$\begin{aligned} P_{\text{high}} &= \{p_i \in P | s_i \geq \tau_{\text{high}}\} \\ P_{\text{unc}} &= \{p_i \in P | \tau_{\text{low}} < s_i < \tau_{\text{high}}\} \\ P_{\text{low}} &= \{p_i \in P | s_i \leq \tau_{\text{low}}\} \end{aligned} \tag{2}$$

where τ_{high} is the threshold for high-quality predictions and τ_{low} is the threshold to separate uncertain predictions and low-quality predictions. After that, we dynamically adjust the image I to obtain the new view I'' based on the feedback from the positional information in the above hierarchical prediction sets. This includes applying a masking operation to the image based on low-confidence predictions and a replacement operation based on uncertain predictions. Specifically, we define

$$I' = \mathbb{M}(I, P_{\text{low}}), \quad I'' = \mathbb{R}\left(I', P_{\text{unc}}\right) \tag{3}$$

where $\mathbb{M}(\cdot)$ and $\mathbb{R}(\cdot)$ represent the *masking* and *pseudo-sample replacement* operations, respectively, applied to the image based on the given prediction set, which will be explained in detail in Section III-D. Finally, the pseudo-labels set Y_p'' for the new view I'' is obtained as $Y_p'' = P_{\text{high}} \cup P_{\text{unc}}$.

Through the pseudo-sample replacement and masking operations on the image, our HPS eliminates the risk of treating uncertain predictions as negative samples and enhances the training process with more robust supervision.

3) Prediction Consistency Learning → MLC Learning: In the aforementioned mainstream pipeline, consistency regularization is implicitly embedded in the output space through the requirement of prediction consistency across different augmented views. However, this prediction consistency constraint heavily depends on the quantity and quality of pseudo-labels, where an insufficient quantity weakens the constraint and poor quality amplifies supervision errors. Moreover, it operates at a high, semantic level, overlooking the learning of fine-grained features. To address these limitations, we propose an MLC learning that simultaneously leverages consistency across multiple levels, including prediction and feature consistency. Building on basic prediction consistency, our method incorporates feature-level consistency, focusing on the features of regions that are potentially foreground. This enhances the learning of fine-grained features and facilitates the exploration of additional potential object regions, providing a richer and more comprehensive supervisory signal.

4) Dynamic Multiview Learning: Building upon HPS and MLC, we propose DML, illustrated in Fig. 4. In DML, for a given image I, weak and strong augmentations are applied to obtain two different views: I_{v1} and I_{v2}. The teacher network performs forward propagation about I_{v1} to generate the predictions. These predictions are then separated by HPS and fed back into the strongly augmented view I_{v2}, generating an updated refined view I_{v3} and its corresponding pseudo-label

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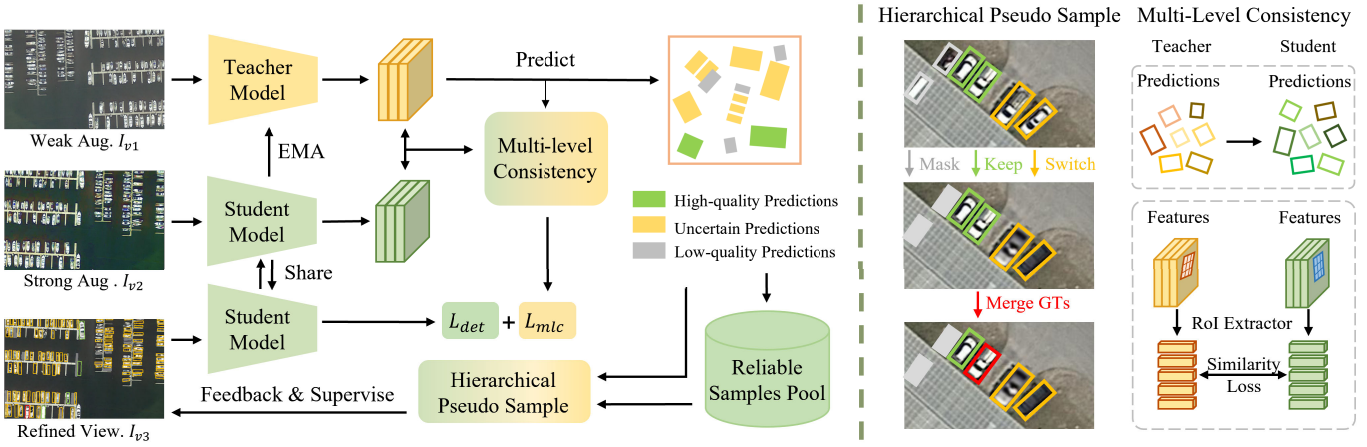


Fig. 4. Pipeline of the proposed DML, based on the T–S architecture, integrates (right) HPS learning and MLC learning. The "reliable samples pool" in the figure is populated with existing sparse annotations and high-quality predictions obtained throughout the training process.

set Y_p. The student network is trained using two components: a) the refined view I_{v3} is used to train the detector, supervised by the union of the pseudo-labels Y_p and sparse annotations G^k, calculating the detection loss L_{det} and b) additionally, using MLC, the consistency loss L_{mlc} between the feature representations of I_{v2} and I_{v3} is computed by leveraging the dense predictions of the teacher network. To make the whole process easier to understand, we also provide the pseudo-code, as shown in Algorithm 1. The total training loss L is defined as follows:

$$L = L_{\text{det}} + \alpha L_{\text{mlc}} \tag{4}$$

where L_{det} is the detection loss of the adopted base detector in the T–S architecture and α is the weight-controlling contribution of MLC loss.

D. HPS Learning

To begin with, the teacher network performs forward propagation on the weakly augmented view I_{v1}, producing a prediction set P. HPS divides P into three sets based on confidence thresholds τ_{high} and τ_{low}: high-quality predictions P_{high}, uncertain predictions P_{unc}, and low-quality predictions P_{low}.

For *high-quality predictions* P_{high}, we retain them as reliable pseudo-labels. *Uncertain predictions* P_{unc} often mix foreground objects with background elements that have similar features. Using these directly as pseudo-labels may introduce false positives (FPs) while discarding them as negative samples could miss actual objects. To address this, we replace these uncertain objects in the image with trustworthy objects of similar size and aspect ratio from the *reliable samples pool*, a process defined as $\mathbb{R}(\cdot)$. This replacement operation improves the reliability of the uncertain predictions, enabling them to be used as pseudo-labels for supervision. *Low-quality predictions* P_{low} typically contain numerous incorrect samples and are treated as negative samples during training. However, due to foreground–background confusion, some real objects may be inadvertently included. To mitigate this, we apply a masking operation $\mathbb{M}(\cdot)$ to these regions in the image, assigning them a gray value of 128. This masking operation ensures these low-quality predictions are treated as true backgrounds, reducing confusion during training.

In implementation, we use the sparsely annotated samples and high-quality prediction samples generated during training to fill the *reliable samples pool* for the replacement operation

Algorithm 1 Dynamic Multiview Learning

Preprocessing:
Train an initial detector $A(\cdot)$ using sparse annotations and images
Initialize teacher $T(\cdot)$ and student $S(\cdot)$ networks with weights from $A(\cdot)$
Main Loop:
for each iteration **do**
  Input image I with sparse annotations G^k
  Generate weakly augmented view I_{v1} and strongly augmented view I_{v2}
  –Teacher Network Forward Pass:
  Obtain dense proposals S_{v1} and the corresponding ROI features $F_{S_{v1}}$
  Generate final predictions $P = T(I_{v1})$
  Categorize predictions based on confidence scores s:
  High-quality predictions P_{high}: $s \geq \tau_{high}$
  Uncertain predictions P_{unc}: $\tau_{low} \leq s < \tau_{high}$
  Low-quality predictions P_{low}: $s < \tau_{low}$
  –View Refinement:
  Generate refined view I_{v3} by switching and masking in I_{v2} based on P_{unc} and P_{low}
  –Student Network Training:
  Extract features from I_{v2} and obtain ROI features $F_{S_{v2}}$ using proposals from S_{v1}
  Compute MLC loss L_{mlc} between $F_{S_{v1}}$ and $F_{S_{v2}}$
  Train $S(\cdot)$ on I_{v3} with mapped $(P_{high} \cup P_{unc} \cup G^k)$, compute detection loss L_{det}
  –Parameter Update:
  $T(\cdot) \xleftarrow{\text{EMA}} S(\cdot)$
end for

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$\mathbb{R}(\cdot)$. When selecting similar samples, we identify objects in the reliable sample pool whose scales and aspect ratios fall within 90%–110% of the target object's values as candidates. One candidate is then randomly selected for patch replacement, ensuring randomness in the replacement process.

E. MLC Learning

Due to the sparse annotations, prediction consistency learning in the output space of the vanilla T–S pipeline tends to be limited and insufficient because of the scarcity of high-quality pseudo-labels. To address this, we propose MLC learning. In addition to the basic consistency regularization in the output space, we introduce dense proposals generated by the teacher network during forward propagation to facilitate dense feature consistency learning in the feature space. In two-stage networks, the RPN can produce dense proposals after training, even if these proposals are not entirely accurate. The large number of proposals sufficiently covers as many foreground objects as possible, allowing feature consistency learning to focus on foreground regions and compensating for the limitations of consistency learning in the feature space.

Specifically, the teacher network performs forward propagation on the weakly augmented view I_{v1}, generating a dense proposal set denoted as $S_{v1} = \{b_i | i = 0, 1, 2, \ldots, N_s\}$, b_i is the bounding box of the ith proposal, and N_s is the number of proposals generated by teacher network. Using the affine transformation relationship between the weak and strong augmentation, we map this dense proposal set to the strong augmented view I_{v2}, obtaining S_{v2}. Based on the proposals in S_{v1} and S_{v2}, we extract the RoI features from the feature maps F_{v1} (from the teacher network for I_{v1}) and F_{v2} (from the student network for I_{v2}), denoted as $F_{S_{v1}}$ and $F_{S_{v2}}$, respectively. Although different augmentations are applied, the semantic features of these potential foreground regions remain consistent. Therefore, we calculate the feature consistency loss L_{mlc} to conduct dense feature consistency learning. We use a cosine similarity loss, formulated as follows:

$$F_{S_{v1}} = T(F_{v1}, S_{v1}), \quad F_{S_{v2}} = S(F_{v2}, S_{v2}) \tag{5}$$

$$L_{\text{mlc}} = 1 - \text{Sim}(F_{S_{v1}}, F_{S_{v2}}) \tag{6}$$

where $T(\cdot)$ and $S(\cdot)$ denote the RoI feature extractors of the teacher and student networks, respectively. $\text{Sim}(\cdot)$ represents the cosine similarity function.

IV. Experiments

A. Datasets

To validate the effectiveness of our method, we conduct experiments on several representative aerial image object detection datasets, including the multiscale and multiclass datasets DOTA-v1.0 [1] and DOTA-v2.0 [7], the small object detection dataset SODA-A [67], and the fine-grained object detection dataset FAIR1M [68]. Additionally, to assess the generalizability of our approach across different domains, we evaluate it on the densely packed object detection dataset SKU-110K-R [69].

TABLE I
Annotation Ratios of Each Dataset Under the Proposed K-ALOD Labeling Pattern

Dataset	objs/img	classes	K=1	K=2	K=3
DOTA-v1.0 [1]	67.10	15	7.78%	13.38%	17.92%
DOTA-v2.0 [7]	159.18	18	5.20%	8.96%	12.05%
SODA-A [67]	347.02	9	5.47%	9.92%	13.68%
FAIR1M-v2 [68]	23.83	37	13.19%	21.22%	27.08%
SKU-110K-R [69]	147.04	1	0.68%	1.36%	2.04%

1) DOTA-v1.0 [1]: DOTA-v1.0 is one of the most popular large-scale aerial image object detection datasets, comprising 15 categories and 2806 images with sizes ranging from 800 × 800 to 20 000 × 20 000 and containing over 188 000 objects. Some categories, such as "*ship*" and "*small-vehicle*," have a large number of densely distributed objects.

2) DOTA-v2.0 [7]: DOTA-v2.0 is an expanded version of DOTA-v1.0, including more aerial images and additional annotations for small objects. It contains 18 categories, 11 268 images, and 1 793 658 objects. Each image contains an average of 159 objects, with many densely distributed scenes.

3) SODA-A [67]: SODA-A is an aerial image dataset for small-object detection. It includes 2513 images and 872 069 objects across nine categories, averaging 347 objects per image. The instance distribution in SODA-A is even denser compared to DOTA-v2.0.

4) FAIR1M [68]: FAIR1M dataset is a large-scale, fine-grained object detection dataset for aerial images, comprising over 40 000 images and more than one million instances. It contains 37 subdivided categories that fall into five major categories: airplanes, ships, vehicles, courts, and roads. Here, we choose FAIR1M-v2 which is supplemented with the validation set for experiments.

5) SKU-110K-R [69]: SKU-110K-R is tailed for detecting densely packed and oriented objects. It contains over 11 000 images of supermarket shelves and more than 1.7 million instance objects. These images are captured from thousands of supermarkets, featuring various scales, perspectives, lighting conditions, and noise levels. Most of the objects in the dataset are tightly packed together.

For the datasets above, we simulate K-ALOD annotations following the above-mentioned rules. K represents the number of selected annotation bases. If the object quantity of any category in an image is less than K, take the maximum available amount. (Due to the large width of aerial images, the input images of training are regarded as the sampling unit). In this study, we experiment with $K = 1, 2, 3$ to verify the effectiveness of the proposed label pattern and learning methods. Table I shows the average number of objects per image (objs/img), the total number of categories, and the labeled objects ratio R for each dataset under the K-ALOD labeling pattern. The labeled objects ratio is calculated through $R = N_{\text{labeled}}/N_{\text{all}}$, where N_{labeled} denotes the total number of labeled objects and N_{all} represents the total number of all objects in the dataset. As shown in Table I, the first four aerial image datasets exhibit a high density of objects per image. In this context, the K-ALOD sampling method demonstrates

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its effectiveness in significantly reducing annotation costs, as it requires labeling only a minimal proportion of object instances. For instance, under the 1-ALOD configuration, it requires labeling less than 15% of the total objects, achieving at least an 85% reduction in annotation costs across these widely used aerial image datasets. Importantly, this efficiency is achieved while maintaining at least 75% of fully supervised performance, as evidenced by the subsequent experimental results. This reduction in labeling costs is particularly significant for extremely dense scenarios in the SKU-110K-R dataset, where the 1-ALOD labeling pattern can reduce annotation costs by more than two orders of magnitude. During training, the sparse annotations are used to supervise the learning process. To ensure accurate evaluation, testing is conducted on fully annotated data.

B. Implementations Details

To facilitate comparison with previous methods, we use the classic and representative two-stage detector Rotated Faster R-CNN (a rotated variant of Faster R-CNN [70]) as our baseline detector. We adopt the ResNet50+FPN structure as the backbone network for all experiments. All experiments are conducted on a single RTX 4090 GPU. Unlike previous methods that directly employ the T–S framework for training, our approach first trains the detector using sparse annotations. The pretrained model is then used to initialize the T–S model, followed by simple fine-tuning with the proposed DML method. Specifically, we train the detector with sparse annotations for 200k iterations and then perform DML fine-tuning for an additional 5k iterations. During network training, the batch size is set to 2, and the learning rate is 0.005. We use stochastic gradient descent (SGD) for backpropagation. For data augmentations, the weak augmentation includes random flipping, while the strong augmentation incorporates random resizing, random color adjustments, random contrast changes, and random sharpness modifications.

For aerial image cropping, we follow the default settings of the original datasets. The DOTA-v1.0, DOTA-v2.0, and FAIR1M-v2 datasets are cropped into 1024×1024 patches with a 200-pixel overlap, while the SODA-A dataset is cropped into 800×800 patches with a 150-pixel overlap. The SKU-100K-R dataset retains its default setting of 1333×800.

The detection performance is evaluated using the standard metric for aerial object detection: mean average precision (mAP) at an intersection over union (IoU) threshold of 0.5. For each class, predictions are compared to GT objects to compute the IoU. Predictions with an IoU ≥ 0.5 are considered true positives (TPs), while those with IoU < 0.5 are classified as FPs. Unmatched GT objects are counted as false negatives (FNs). Precision and recall are computed based on the counts of TPs, FPs, and FNs. The average precision (AP) for each class is then calculated as the area under the precision–recall (P–R) curve. Finally, the mAP is obtained by averaging the AP across all classes.

C. Main Results

1) Comparison of Label Efficiency: To compare the label efficiency under different labeling patterns, we evaluate the detection performance of various label-efficient patterns and methods, including SSOD and SAOD, while ensuring the same annotation ratio. Using the 1-ALOD setting on the DOTA-v1.0 and FAIR1M datasets as the baseline, we retain full annotations for the same proportion of images in the SSOD setting, leaving the rest unannotated. We evaluate the performance of mainstream semi-supervised detectors, such as Soft-teacher [39] and the latest rotated object semi-supervised detection method, SOOD [8], on this semi-supervised training data. For SAOD, we randomly retain the same proportion of annotations and compare the results with several state-of-the-art SAOD methods. Note that [66] has proposed the SIOD annotation format, which involves annotating only one instance per category per image. This represents a specific case of our proposed K-ALOD, where $K = 1$. However, the method introduced in [66], DMiner, is limited to 1-ALOD annotations and requires a specially designed network, which limits its generalization capabilities. We also include a comparison with this approach here.

TABLE II
COMPARISON OF SEVERAL DIFFERENT LABEL-EFFICIENT PATTERNS AND TRAINING METHODS. REPORT MAP (%)

Method	Labeling Pattern	DOTA-v1.0	FAIR1M
Baseline	SSOD	52.5	28.0
Baseline	SAOD	49.3	25.6
Baseline	ALOD	**53.3**	**28.7**
Soft-teacher [39]	SSOD	56.9	26.2
SOOD [8]	SSOD	54.4	26.4
Co-mining [61]	SAOD	51.6	23.4
CalibrateTeacher [13]	SAOD	51.7	26.9
DMiner [66]	SIOD	57.2	29.2
DML(Ours)	ALOD	**61.1**	**29.8**

Table II presents the comparison results, where "baseline" refers to the performance of the baseline detector trained directly using the corresponding incomplete annotations. Compared to other random labeling patterns, the proposed ALOD achieves more efficient and better performance. By focusing on category information during random annotation (without significantly increasing annotation costs), ALOD delivers superior detection results. Furthermore, when using various label-efficient methods under their respective training setups, our method consistently outperforms others, demonstrating the effectiveness of both our labeling pattern and training method DML.

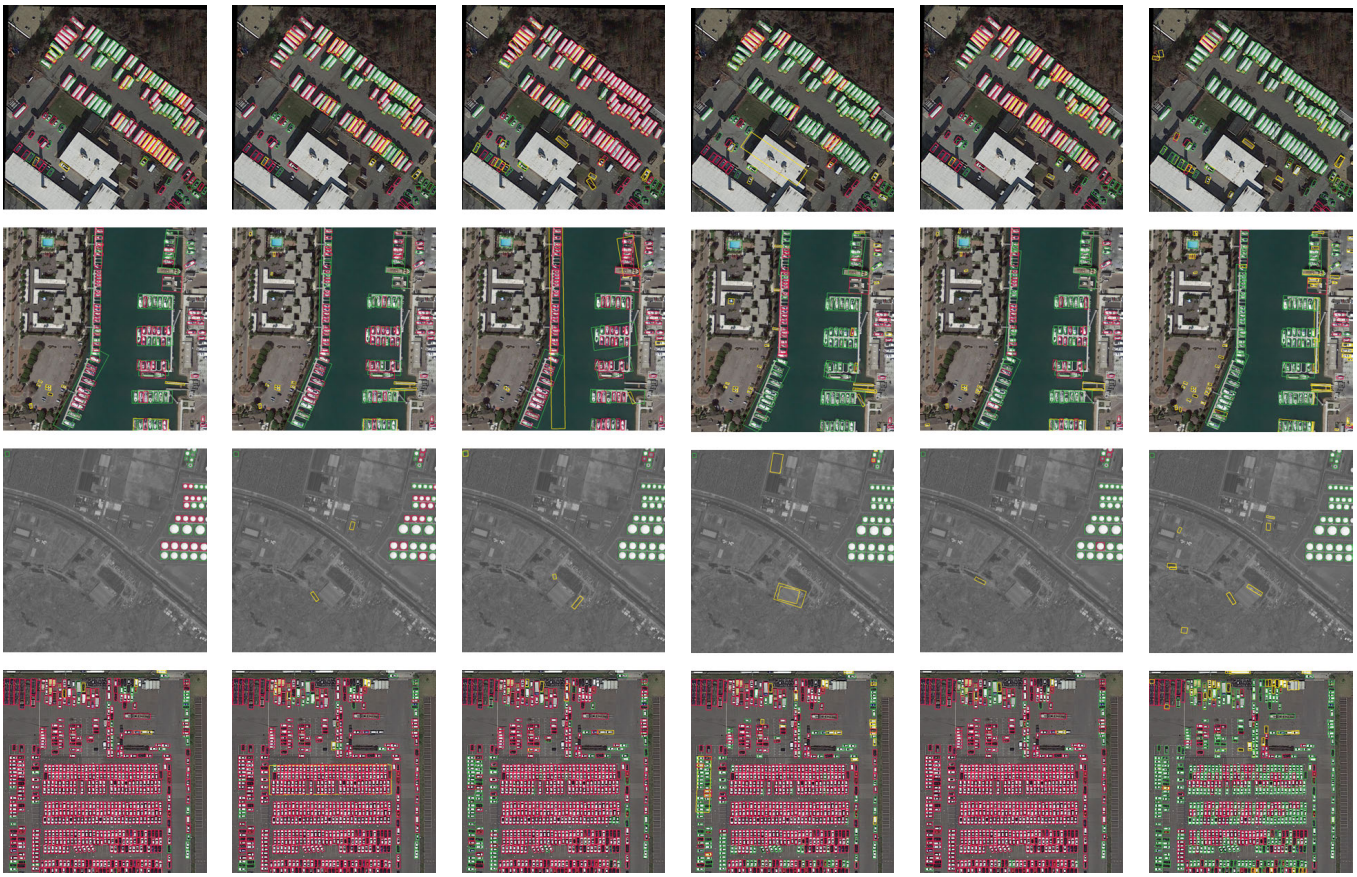
2) Comparison of Learning Strategies: To compare the advantages of the proposed learning strategy, we use the same K-ALOD setting to evaluate the performance of the proposed DML with state-of-the-art SAOD methods (e.g., Co-mining [61], RobustOD [14], SparseDet [12], and Calibrated Teacher [13]). To ensure a fair comparison, we adopt Rotated Faster R-CNN as the baseline detector and re-implement the rotated object detection versions of the aforementioned methods. The results are shown in Table III, where our method achieves the best performance on all datasets. In the table, *Oracle* refers to the results obtained through

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TABLE III

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED DML AND SOTA SAOD METHODS ON SEVERAL AERIAL IMAGE OBJECT DETECTION DATASETS, UNDER THE SAME K-ALOD ANNOTATIONS. ALL METHODS USE THE ROTATED FASTER R-CNN AS THE BASE DETECTOR. "ORACLE" REFERS TO RESULTS FROM TRAINING WITH COMPLETE ANNOTATIONS, WHILE THE OTHER METHODS USE K-ALOD ANNOTATIONS. "VANILLA" REPRESENTS THE RESULTS FROM DIRECTLY TRAINING THE DETECTOR WITH K-ALOD ANNOTATIONS. TO ENSURE A FAIR COMPARISON, THE AUGMENTATION STRATEGIES USED IN THE T–S FRAMEWORK WERE ALSO APPLIED TO THE "ORACLE" AND "VANILLA." REPORT MAP (%)

Method	DOTA-v1.0			DOTA-v2.0			SODA-A			FAIR1M		
	$K=1$	$K=2$	$K=3$	$K=1$	$K=2$	$K=3$	$K=1$	$K=2$	$K=3$	$K=1$	$K=2$	$K=3$
Oracle		70.4			61.9			63.4			35.0	
Vanilla	53.3	62.3	64.2	51.4	54.9	58.1	47.7	52.7	54.3	28.7	31.4	32.8
Co-mining [61]	56.8	62.8	64.3	49.5	54.4	56.4	46.4	50.8	53.2	26.8	27.0	29.1
SparseDet [12]	47.2	54.1	56.1	43.6	48.8	50.2	43.2	49.3	53.9	28.3	30.5	31.0
Robust OD [14]	50.1	54.7	56.9	40.6	47.4	50.3	45.1	49.0	49.9	24.5	26.9	27.7
Calibrated Teacher [13]	55.9	63.1	65.0	50.8	55.3	56.7	47.0	52.9	53.2	29.4	31.6	32.7
DML(Ours)	**61.1** (+7.8)	**63.6** (+1.3)	**65.2** (+1.0)	**55.6** (+4.2)	**57.7** (+2.8)	**59.8** (+1.7)	**54.9** (+7.2)	**55.9** (+3.2)	**56.4** (+2.1)	**29.8** (+1.1)	**32.4** (+1.0)	**33.3** (+0.5)



(a) (b) (c) (d) (e) (f)

Fig. 5. Visualization of the detection results of different methods on the DOTA-v1.0 dataset, under the 1-ALOD setting. Correct predictions are marked with green boxes, FP predictions are marked with yellow boxes, and missing targets are marked with red boxes. (a) Supervised. (b) Co-mining [61]. (c) SparseDet [12]. (d) ROD [14]. (e) CT [13]. (f) DML (ours).

fully supervised learning using all annotations, while *Vanilla* represents the results of supervised learning under ALOD's sparse annotations, with K set to 1, 2, and 3. (Note that we also apply the same strong augmentation in the "*Oracle*" and "*Vanilla*" setting for a fair comparison.) Under the ALOD annotations setting, *Vanilla* multiclass object detection in aerial images can achieve over 75% of full supervision performance with just one annotation per class per image. With 3-ALOD annotations, it can achieve over 90% of full supervision performance. These results demonstrate that our proposed label pattern can significantly reduce annotation costs while minimizing the loss of detection performance.

Furthermore, utilizing the same ALOD annotations, the proposed DML pipeline demonstrates superior performance compared to other SAOD methods. On the multiscale and multiclass aerial image dataset DOTA-v1.0, our method achieves improvements of 7.8, 1.3, and 1.0 mAP (Table III; DOTA-v1.0) over the *Vanilla* baseline under 1/2/3-ALOD annotation conditions, respectively. On the DOTA-v2.0 dataset, which contains massive small objects, some state-of-the-art

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TABLE IV

QUANTITATIVE COMPARISON BETWEEN THE PROPOSED DML AND SOTA SAOD METHODS ON THE SKU-110K-R DATASET. REPORT MAP (%)

Method	$K = 1$	$K = 2$	$K = 3$
Oracle		79.9	
Vanilla	17.0	57.2	63.5
Co-mining [61]	16.9	56.7	64.3
Robust OD [14]	24.0	56.9	64.8
Calibrated Teacher [13]	17.2	58.0	65.8
DML (Ours)	**49.7** (+32.7)	**67.3** (+10.1)	**68.0** (+4.5)

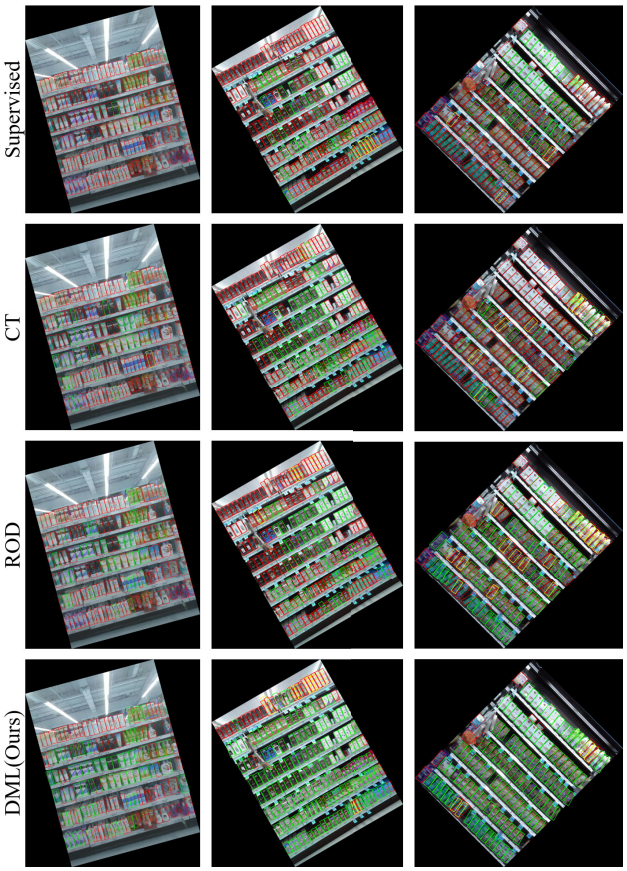


Fig. 6. Visualization of the detection results of different methods on the SKU-100K-R [69] dataset, under the 1-ALOD setting (the SKU-110K-R dataset is the rotated version of the original SKU-110K dataset). Correct predictions are marked with green boxes, FP predictions are marked with yellow boxes, and missing targets are marked with red boxes.

SAOD methods seem to falter. However, our method still surpasses the *Vanilla* baseline with mAP improvements of 4.2, 2.8, and 1.7 (Table III; DOTA-v2.0) under 1/2/3-ALOD annotation conditions, respectively. Similarly, our method demonstrates superior performance on the small-object dataset SODA-A and the fine-grained object dataset FAIR1M (Table III; SODA-A and FAIR1M), consistently outperforming other methods.

3) Visualization: Fig. 5 presents the visualization results of our proposed DML method compared to other SOTA approaches on the DOTA-v1.0 dataset. From the figure, it is evident that our method demonstrates superior detection performance across various categories. Notably, our DML excels

TABLE V

COMPONENT ANALYSIS OF OUR PROPOSED METHOD. THE T–S MODEL MEANS USING THE TEACHER–STUDENT ARCHITECTURE

	T-S model	HPS	MCL	mAP(%)
Baseline				53.3
Different Configurations	✓			55.3
	✓	✓		60.6
	✓		✓	56.8
	✓	✓	✓	**61.1**

TABLE VI

PERFORMANCE COMPARISON UNDER DIFFERENT α VALUES

α	0.5	1.0	2.0	3.0	4.0
mAP(%)	60.5	**61.1**	60.6	60.2	60.0

in detecting densely packed objects, as demonstrated by the "small-vehicle" category in the last row of Fig. 5. Under the 1-ALOD label pattern, where only one annotated instance is retained per densely packed object, the supervision information becomes critically sparse. Conventional methods often struggle to learn effectively from such sparse supervision due to the limited number of annotated instances. In contrast, our DML overcomes this challenge through multiview learning, which provides richer and more reliable supervised information. This mechanism significantly enhances the localization accuracy in visually complex regions, resulting in more compelling visual results.

4) Generalization Ability: The proposed labeling pattern and learning method aims to alleviate the issue of redundant annotations for densely packed objects in aerial imagery. To validate the generalization performance of our label and method, we conducted comparative experiments on a densely packed object detection dataset. Specifically, we applied the 1/2/3-ALOD annotation method to the SKU-110K-R dataset. The results, shown in Table IV, compare our approach to other SOTA SAOD methods. From the results, it is clear that even for densely packed objects in general images, our method achieves detection performance close to fully annotated models while significantly reducing the number of annotations needed.

Similarly, we visualize the results of the SKU-110K-R dataset under the 1-ALOD setting, as shown in Fig. 6. In SKU-110K-R, each image contains numerous objects, but under the 1-ALOD annotation setting, only one instance per image is annotated. Despite the low annotation ratio, our method can generate more reliable supervision information compared to other methods, achieving superior results.

D. Ablations

1) Component Analysis: We evaluate the effectiveness of each module proposed in this article through a series of ablation experiments. Unless otherwise stated, all ablation experiments are conducted on the DOTA-v1.0 dataset using the 1-ALOD labeling pattern to ensure uniformity and fair

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TABLE VII
ABLATION STUDIES AND HYPERPARAMETER SELECTION IN THE DESIGNED MODULES. TRAINING ON THE 1-ALOD LABELED DOTA-V1.0

(a)

	switch	mask	mAP(%)
T-S finetune			55.3
T-S finetune with operations	✓		60.1
		✓	56.6
	✓	✓	**60.6**

(b)

τ_h	0.6	0.7	0.8	0.9
mAP(%)	60.1	60.1	60.8	**61.1**
τ_l	0	0.1	0.2	0.3
mAP(%)	60.0	60.7	**61.1**	60.5

(c)

Num	Features	mAP(%)
I	All	59.7
II	Dense Prediction	**61.1**
III	Final Prediction	60.2



Fig. 7. Visualization of the refined view I_{v3} after applying the HPS module. The red boxes represent the original sparse label GT, the green boxes represent the remaining high-quality pseudo-labels, the yellow boxes represent the boxes with the switch operation applied, and the gray boxes represent the mask areas.

comparison. The results are shown in Table V. The baseline detector, Rotated Faster RCNN, achieves 53.3% mAP with 1-ALOD annotations. After introducing the T–S model, the simplest high-confidence pseudo-label mining approach further improves detection performance to 55.7%. With the stepwise introduction of the proposed HPS and MCL modules, the results show progressive mAP improvements, demonstrating the effectiveness of each designed module. Among them, HPS provides the most significant improvement, as it directly introduces more effective positive samples for training the detector. We also visualized the refined views produced by HPS, as shown in Fig. 7. Compared to traditional high-confidence threshold pseudo-labeling, HPS supplements a richer set of effective training samples.

We also conduct an experimental analysis of the parameter α, which controls the loss weight of the MLC module in the overall loss function. The experimental results under different values of α are presented in Table VI. The results indicate that the optimal performance is achieved when $\alpha = 1.0$.

2) Operations and Hyperparameter in HPS: Compared to traditional binary pseudo-labeling learning, HPS incorporates information from the teacher's predictions to refine the input image, primarily including the *switch* operation for uncertain predictions and the *mask* operation for low-quality predictions. We conduct ablation experiments on these two operations, and the results are shown in Table VII(a). It can be observed that both operations positively impact object detection under sparse annotations: switch introduces more reliable targets for training, while mask suppresses confusion in the unlabeled regions.

Two parameters, τ_l and τ_h, are involved in categorizing teacher predictions. We compare multiple settings for these parameters, and the results are summarized in Table VII(b). The optimal values are determined to be $\tau_l = 0.2$ and $\tau_h = 0.9$.

3) Different Feature Consistency Strategies in MLC: In MLC learning, we utilize the dense predictions generated by the teacher model to further align features across different views, thereby enhancing consistency learning. In a two-stage detector, the dense predictions are derived from the outputs of the RPN. These outputs effectively balance the exploration of potential foreground regions while simultaneously filtering out many irrelevant background areas. Here, we conduct a comparative analysis of three different features used for consistency learning.

1) *All Features:* all the feature maps extracted by the backbone network.
2) *Features of Dense Prediction:* the RoI features of dense predictions generated by the RPN.
3) *Features of Final Prediction:* the RoI features of all the final predictions.

The results, as presented in Table VII(c), indicate that the feature consistency based on dense predictions yields superior performance.

V. DISCUSSION

Although the proposed ALOD label pattern also employs sparse annotations, it differs from previous SAOD methods by introducing constraints on class labels rather than relying on random sparse annotations. As a result, ALOD helps mitigate class imbalance issues, particularly when dealing with real-world, unbalanced data. For example, in DOTA-v2.0, where "small-vehicle" constitutes over 30% of the dataset and often appears in clusters, ALOD ensures more equitable treatment of each class. This approach alleviates the impact of unbalanced class distributions and improves the overall effectiveness of the training data. Furthermore, compared to SSOD, which densely annotates all objects in a small subset of images, retaining the original class imbalance and suffering from domain gap issues, ALOD offers a more balanced and effective solution.

While our proposed ALOD achieves near fully supervised detection performance with minimal annotation efforts, several limitations warrant further discussion. First, under sparse annotation conditions, we inevitably miss numerous challenging samples that are crucial for model training. These hard cases are particularly difficult to distinguish and effectively utilize as accurate pseudo-labels during the training process, resulting in a performance gap compared to fully supervised methods. This limitation, we note, is inherent to

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all label-efficient approaches and represents a fundamental challenge in the field. To address this issue, future research directions could explore the integration of few-shot learning techniques with the powerful representation capabilities of large foundation models, potentially enabling more effective mining and utilization of challenging samples in unannotated regions. Second, while our method demonstrates promising results, the performance gap between sparse annotation and full supervision still exists. However, empirical observations reveal that a substantial proportion of instances can be accurately predicted through relatively simple network training. This insight suggests promising future research directions in optimizing annotation strategies to potentially achieve fully supervised performance levels, particularly through the development of more sophisticated sample selection mechanisms and adaptive annotation patterns.

VI. Conclusion

In this work, we have presented a label-efficient object detection pipeline for aerial imagery. Inspired by aerial images' unique imaging properties and object attributes, we propose a novel and efficient attribute-aware labeling pattern that samples K class-orthogonal annotation bases for each image. This pattern minimizes sample redundancy, retains a more diverse set of samples, and significantly enhances label utility. Furthermore, we develop a DML strategy to address the challenges posed by sparse annotations. To improve data utility, we further introduce HPS learning by separating model predictions into multiple hierarchies, which guides the dynamic construction of new views to expand sample quantity while ensuring sample quality. Additionally, we propose MLC learning, aligning both low-level features and high-level predictions to guarantee robust and faster label-efficient learning. Extensive experiments on five benchmark datasets validate the effectiveness of our method, demonstrating state-of-the-art accuracy with minimal sample redundancy.

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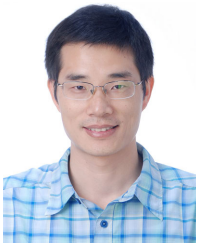
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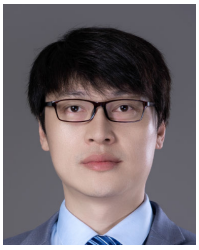
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